

LexVert — Environment Variables Setup Checklist

Everything that needs to be filled in before the app runs, pulled directly from the code that reads each variable. Two separate env files: `.env.local` at the repo root (Next.js frontend) and `backend/.env` (FastAPI + RAG service).

How this maps to files:

•

`.env.example` (repo root) → copy to `.env.local` — frontend variables below.

•

`backend/.env.example` → copy to `backend/.env` — backend/RAG variables below.

REQUIRED

app breaks or a whole feature 500s without it

RECOMMENDED

works without it, but degraded/unsafe in production

OPTIONAL

feature-gated,

has a safe fallback

SCAFFOLDED

no code reads it yet

Frontend — .env.local

repo root

Core / App

VARIABLE	STATUS	WHERE TO GET IT	NOTES
☐ NEXT_PUBLIC_APP_URL	REQUIRED	Your deployed frontend URL	Used in checkout redirects, email links, CSRF origin check
☐ NEXT_PUBLIC_BACKEND_API	REQUIRED	Your FastAPI backend URL	<code>NEXT_PUBLIC_BACKEND_URL</code> works too as a fallback name
☐ NODE_ENV	REQUIRED	<code>development</code> / <code>production</code>	Gates CSRF strictness, CORS defaults
☐ NEXT_PUBLIC_APP_ENV	OPTIONAL	free text	Display/label only
☐ BACKEND_INTERNAL_TOKEN	REQUIRED	Generate a random hex token	Must exactly match <code>LEXVERT_INTERNAL_TOKEN</code> in <code>backend/.env</code>
☐ LEXVERT_CORS_ORIGINS	REQUIRED	Comma-separated origin list	Origins the backend accepts requests from
☐ JWT_SECRET	REQUIRED	Random string, 32+ chars	Signs admin-auth JWTs
☐ STRICT_CSRF_PROTECTION	OPTIONAL	<code>true</code> / <code>false</code>	Prod enforces CSRF regardless; use to force it on in dev
☐ SUPPORT_TEAM_EMAIL	REQUIRED	Your support inbox address	Used by complaints + fraud-report routes

Authentication — Clerk

VARIABLE	STATUS	WHERE TO GET IT	NOTES
☐ NEXT_PUBLIC_CLERK_PUBLISHABLE_KEY	REQUIRED	Clerk Dashboard → API Keys	Whole app is behind Clerk auth
☐ CLERK_SECRET_KEY	REQUIRED	Clerk Dashboard → API Keys	
☐ CLERK_WEBHOOK_SECRET	REQUIRED	Clerk Dashboard → Webhooks → Signing Secret	Verifies user-sync webhook calls

Database — MongoDB

VARIABLE	STATUS	WHERE TO GET IT	NOTES
☐ MONGODB_URI	REQUIRED	MongoDB Atlas → Connect → Drivers	
☐ MONGODB_DB	REQUIRED	Your database name	Default <code>LexVert</code>
☐ MONGO_SYNC_INDEXES	OPTIONAL	<code>true</code> once, then unset	Rebuilds indexes on boot

Email — SendGrid & hearing reminders

VARIABLE	STATUS	WHERE TO GET IT	NOTES
☐ SENDGRID_API_KEY	REQUIRED	SendGrid → Settings → API Keys	
☐ SENDGRID_FROM_EMAIL	REQUIRED	A verified sender in SendGrid	

<code>SENDGRID_FROM_NAME</code>	OPTIONAL	free text	Default "LexVert"
<code>NOTIFICATION_FROM_EMAIL</code>	REQUIRED	Same verified sender	
<code>NOTIFICATION_FROM_NAME</code>	OPTIONAL	free text	
<code>NOTIFICATION_REMINDER_WINDOWS_DAYS</code>	OPTIONAL	e.g. 7,3,1	Days-before-hearing to remind
<code>NOTIFICATION_MAX_RETRIES</code>	OPTIONAL	integer	Default 3
<code>ENABLE_HEARING_NOTIFICATIONS</code>	OPTIONAL	true / false	Feature toggle
<code>ENABLE_NOTIFICATION_EMAILS</code>	OPTIONAL	true / false	Feature toggle
<code>CRON_SECRET</code>	REQUIRED	Generate a random hex token	Protects the internal notification-run endpoint

File uploads — Cloudinary

VARIABLE	STATUS	WHERE TO GET IT	NOTES
<code>CLOUDINARY_CLOUD_NAME</code>	REQUIRED	Cloudinary Console → Dashboard	Backs the active file/image upload routes
<code>CLOUDINARY_API_KEY</code>	REQUIRED	Cloudinary Console → Dashboard	
<code>CLOUDINARY_API_SECRET</code>	REQUIRED	Cloudinary Console → Dashboard	

Payments — Stripe

VARIABLE	STATUS	WHERE TO GET IT	NOTES
<code>STRIPE_SECRET_KEY</code>	REQUIRED	Stripe Dashboard → Developers → API Keys	Billing routes no-op (return null client) without it
<code>STRIPE_WEBHOOK_SECRET</code>	REQUIRED	Stripe Dashboard → Developers → Webhooks	
<code>STRIPE_PRICE_STARTER_MONTHLY</code>	REQUIRED	Stripe → Product Catalog → Price ID	
<code>STRIPE_PRICE_STARTER_YEARLY</code>	REQUIRED	Stripe → Product Catalog → Price ID	
<code>STRIPE_PRICE_PROFESSIONAL_MONTHLY</code>	REQUIRED	Stripe → Product Catalog → Price ID	
<code>STRIPE_PRICE_PROFESSIONAL_YEARLY</code>	REQUIRED	Stripe → Product Catalog → Price ID	
<code>STRIPE_PRICE_ENTERPRISE_MONTHLY</code>	REQUIRED	Stripe → Product Catalog → Price ID	
<code>STRIPE_PRICE_ENTERPRISE_YEARLY</code>	REQUIRED	Stripe → Product Catalog → Price ID	

Payments — Razorpay (alternative gateway to Stripe)

VARIABLE	STATUS	WHERE TO GET IT	NOTES
<code>RAZORPAY_KEY_ID</code>	OPTIONAL	Razorpay Dashboard → API Keys	Billing falls back gracefully without it
<code>RAZORPAY_KEY_SECRET</code>	OPTIONAL	Razorpay Dashboard → API Keys	
<code>RAZORPAY_WEBHOOK_SECRET</code>	OPTIONAL	Razorpay Dashboard → Webhooks	
<code>RAZORPAY_AMOUNT_*</code> (×6, per plan/cycle)	OPTIONAL	Your pricing, in paise	STARTER / PROFESSIONAL / ENTERPRISE × MONTHLY / YEARLY

Rate limiting — Upstash Redis

VARIABLE	STATUS	WHERE TO GET IT	NOTES
<code>UPSTASH_REDIS_REST_URL</code>	RECOMMENDED	Upstash Console → Redis DB → REST API	Without it, rate limiting is per-instance in-memory — breaks on serverless (Vercel)
<code>UPSTASH_REDIS_REST_TOKEN</code>	RECOMMENDED	Upstash Console → Redis DB → REST API	

Scaffolded, not wired into any route yet

VARIABLE	STATUS	WHERE TO GET IT	NOTES
TWILIO_ACCOUNT_SID / AUTH_TOKEN / PHONE_NUMBER	SCAFFOLDED	Twilio Console	SMS — no code reads these yet
NEXT_PUBLIC_FIREBASE_* / FIREBASE_SERVICE_ACCOUNT_JSON	SCAFFOLDED	Firebase Console	Push notifications — not integrated
AWS_ACCESS_KEY_ID / SECRET / S3_BUCKET_NAME / REGION	SCAFFOLDED	AWS IAM Console	Cloudinary is the active uploader instead

Backend — backend/.env

FastAPI + RAG

Server & core

VARIABLE	STATUS	WHERE TO GET IT	NOTES
LEXVERT_HOST	OPTIONAL	—	Default 0.0.0.0
PORT / LEXVERT_PORT	OPTIONAL	—	PORT wins (Render/Heroku style); default 8000
LEXVERT_DEBUG	OPTIONAL	true / false	Relaxes CORS/host checks — dev only
ENABLE_SCRAPER_SCHEDULER	OPTIONAL	true / false	
LEXVERT_UPLOAD_DIR	OPTIONAL	—	Default ./uploads
LEXVERT_MAX_PDF_SIZE_MB	OPTIONAL	—	Default 50

Database

VARIABLE	STATUS	WHERE TO GET IT	NOTES
MONGODB_URI	REQUIRED	Same MongoDB Atlas cluster as frontend	
MONGODB_DB	OPTIONAL	—	Default cause_list_db

CORS & security hardening

VARIABLE	STATUS	WHERE TO GET IT	NOTES
LEXVERT_CORS_ORIGINS	REQUIRED in prod	Your deployed frontend URL(s)	Throws on boot in prod if unset
LEXVERT_TRUSTED_HOSTS	OPTIONAL	Your backend's own domain(s)	
LEXVERT_RATE_LIMIT_WINDOW_SECONDS	OPTIONAL	—	Default 60
LEXVERT_RATE_LIMIT_MAX_REQUESTS	OPTIONAL	—	Default 120
LEXVERT_MAX_CONCURRENT_BULK_IMPORTS	OPTIONAL	—	Default 1
LEXVERT_IMPORT_PROGRESS_TTL_SECONDS	OPTIONAL	—	Default 86400, min 300
LEXVERT_INTERNAL_TOKEN	REQUIRED	python -c "import secrets; print(secrets.token_hex(32))"	Must match BACKEND_INTERNAL_TOKEN in frontend .env.local

Auth — Clerk JWT verification

VARIABLE	STATUS	WHERE TO GET IT	NOTES
CLERK_JWT_ISSUER	RECOMMENDED	Clerk Dashboard → API Keys (issuer URL)	Without it, the issuer check is skipped entirely
CLERK_JWT_AUDIENCE	OPTIONAL	Clerk Dashboard	Only enforced if set

Scraper

VARIABLE	STATUS	WHERE TO GET IT	NOTES
<code>PDF_DOWNLOAD_ENABLED</code>	OPTIONAL	true / false	Gates the daily cause-list scrape
<code>COURT_WEBSITE_URL</code>	OPTIONAL	—	Defaults to the Delhi HC cause-list page

RAG ingestion — OpenAI & Chroma

VARIABLE	STATUS	WHERE TO GET IT	NOTES
<code>OPENAI_API_KEY</code>	REQUIRED for RAG	platform.openai.com → API Keys	Powers embeddings, chunking & metadata extraction — nothing in <code>rag/</code> works without it
(no env var) — Chroma	n/a	—	Local on-disk store at <code>rag/data/chroma</code> ; only needs env vars if later switched to a hosted Chroma server

RAG ingestion — Cloudflare R2

VARIABLE	STATUS	WHERE TO GET IT	NOTES
<code>R2_ACCOUNT_ID</code>	REQUIRED for RAG	R2 dashboard → subdomain of your S3 API endpoint	
<code>R2_ACCESS_KEY_ID</code>	REQUIRED for RAG	R2 dashboard → API Token (Object Read & Write, scoped to bucket)	
<code>R2_SECRET_ACCESS_KEY</code>	REQUIRED for RAG	Same R2 API token	
<code>R2_BUCKET</code>	OPTIONAL	—	Default <code>lexvert-docs</code>
<code>R2_ENDPOINT</code>	OPTIONAL	—	Derived from account ID if unset
<code>R2_HASH_BACKUP_KEY</code>	OPTIONAL	—	Ingested-doc hash index backup; no-ops silently if unset
<code>R2_SCRAPE_HASH_BACKUP_KEY</code>	OPTIONAL	—	Separate key for the scraper's "downloaded" index
<code>HASH_DB_MAP_SIZE_MB</code>	OPTIONAL	—	Default 4096
<code>HASH_DB_BACKUP_INTERVAL_SECONDS</code>	OPTIONAL	—	Default 300

RAG ingestion — Docling extraction

VARIABLE	STATUS	WHERE TO GET IT	NOTES
<code>RAG_DOCLING_OCR_FALLBACK</code>	OPTIONAL	true / false	Only retries with OCR on scanned docs; default true
<code>RAG_DOCLING_TABLES</code>	OPTIONAL	true / false	Default true
<code>RAG_DOCLING_LAYOUT_ENGINE</code>	OPTIONAL	<code>onnxruntime</code> or <code>transformers</code>	Keep <code>onnxruntime</code> on Windows — torch segfaults

Generated from an audit of every `os.getenv` / `process.env` read in the codebase (backend/.env site-packages excluded). Real secret values already sitting in `.env.local` / `backend/.env` were not reproduced here — only variable names and where each one comes from.